

Data Analyst Interview Q&A;

Day 7 of 7 — General Interview Practice Set

How to Use This Guide

These are common Data Analyst interview questions, framed around this project so your answers sound specific rather than generic.

1. Walk me through this project.

See [Project_Explanation.pdf](#) for the full 2-minute script — in short: AromaKart, a fictional Indian perfume retailer, needed to understand revenue drivers, seasonality, and returns. I built a 3,000-row synthetic dataset and analyzed it end-to-end across Excel, SQL, Python, and Tableau, landing on 5 concrete business recommendations.

2. Why did you choose to build 4 different tools instead of just one dashboard?

Each tool answers a different kind of question fastest: Excel for quick, stakeholder-shareable summaries; SQL for precise, repeatable aggregations against a real database; Python for deeper exploratory analysis and custom visuals; Tableau for an interactive, presentation-ready dashboard. Using all four also demonstrates I can work in whatever stack a team already has.

3. What was the most interesting insight you found?

That premium perfumes had a meaningfully higher return rate than budget ones — a claim Customer Support had made anecdotally but nobody had measured. Quantifying it turned a hunch into a concrete, actionable finding.

4. How did you validate that your data was clean before analyzing it?

I checked row/column counts, data types, missing values, and duplicate Order_IDs first (a repeatable five-minute checklist), then confirmed Total_Amount was mathematically consistent with Price, Quantity, and Discount_Percentage before trusting any aggregation built on top of it.

5. How would you present these findings to a non-technical stakeholder?

Lead with the business impact, not the method: 'Diwali and New Year drive 2-3x normal order volume, so we should have inventory ready 4-6 weeks earlier' lands better than describing the SUMPRODUCT formula behind it. I'd show one chart per insight, not the full dashboard, to keep the conversation focused.

6. What would you do differently with more time or a bigger team?

Add a cohort/retention analysis to understand how quickly repeat customers churn, build a proper data pipeline (e.g., dbt + a warehouse) instead of a single CSV, and A/B test the discount-reduction recommendation on premium SKUs before rolling it out broadly.

7. How do you decide between a bar chart, line chart, and pie chart?

Line charts for trends over time (revenue by month), bar charts for comparing categories (revenue by city/brand), and pie charts only for simple part-to-whole breakdowns with few categories (payment method mix) — pie charts get unreadable past 5-6 slices, so I'd switch to a bar chart if that mix grew more complex.

8. What's a time you found something in data that contradicted your assumption?

I initially assumed discount depth and order volume would scale together for every brand. The data showed that wasn't true for premium brands — heavy discounting barely moved their volume, which changed my recommendation from 'discount more' to 'discount more selectively.'

9. How do you handle a stakeholder who disagrees with your analysis?

First, I'd check whether they're seeing something in the business I'm missing — data doesn't capture everything. Then I'd walk through the methodology transparently, show the underlying numbers, and if we still disagree, propose a small test to settle it with evidence rather than opinion.

10. Why should we hire a junior analyst instead of someone more senior for this role?

I bring current tool fluency, a genuine habit of validating data before trusting it, and no bad habits to unlearn — plus I'm significantly more cost-effective while I build toward senior-level judgment on the job.